

controlJacobian.m Documentation

controlJacobian.m

Purpose

Computes the geometric Jacobian of the robot.

It maps joint velocities to TCP velocity:

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = J(q)\dot{q}$$

Inputs

- `robot` — robot DH parameters and TCP transform
- `q` — current joint angles

Output

- `J` — 6×6 geometric Jacobian

Main Idea

For each revolute joint:

$$J_{v_i} = z_{i-1} \times (p_{TCP} - p_{i-1})$$
$$J_{\omega_i} = z_{i-1}$$

Therefore:

$$J_i = \begin{bmatrix} J_{v_i} \\ J_{\omega_i} \end{bmatrix}$$

Main Steps

1. Run through the same DH chain used in FK.

2. Before each joint transformation, store:
 - `p_i` — joint origin
 - `z_i` — joint rotation axis
3. Find final TCP position.
4. Build each Jacobian column using the equations above.

Convention Used

`controlJacobian.m` :

```
Rows 1-3 → linear velocity  
Rows 4-6 → angular velocity
```

MATLAB `geometricJacobian()` uses the opposite order:

```
Rows 1-3 → angular velocity  
Rows 4-6 → linear velocity
```

So MATLAB's result must be reordered before validation.

Validation

Validated against MATLAB `geometricJacobian()` .

Difference was around:

$$10^{-16}$$

which is numerical round-off.

Used For

- TCP velocity
- Singularity checks
- Differential IK
- Welding-path velocity analysis

Related Files

- `dhTransform.m`
- `controlFK.m`
- `symbolicJacobian.m`